

Shareholders Conference FY2025

ENTERPRISE ADVANCED TECHNOLOGIES CORP.

Laptop Refresh & IT Asset Management SaaS Platform

INTERNAL AND EXTERNAL LAPTOP REFRESH PROJECT

TEAM: ANH • KELLY MAPE • ALMENDRA SANCHEZ

Today's Speakers

Enterprise **A**dvanced **T**echnologies



Jason Huang Ahhhh

SVP

Software, **H**ardware, **I**T **S**ervices



Almendra Sanchez

EVP

Operations & **F**inance



Kelly Mape

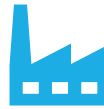
GM

Research & **D**evelopment

Agenda



Introduction &
Problem Context



Current Business
Environment & Market



Development
Methodology



Solution Architecture
& Workflows



System Analysis &
Design



Conclusion & Next
Steps

Introduction & Problem Context

“

Know thyself, know thy enemy

”

SUN TZU, PROBABLY

Know thyself

Our Company & Problem Overview



Enterprise Advanced Technologies – founded 2007



HRM and physical asset management solutions



Current Fleet: 5,000 active + 1,000 backup Lenovo T470
4 waves of hardware refresh



Windows 11 CPU, Security & Compliance requirements → current laptops become obsolete

Current State of Affairs

	C	D	E	F	G
	Employee ID	Email	Manager	Department	Assigned Laptop ID
1	EAT001	emma.johnson@eat.app	Olivia Brown	Engineering	LAP1001
2	EAT002	liam.smith@eat.app	Olivia Brown	Engineering	LAP1002
3	EAT003	olivia.brown@eat.app		Engineering	LAP1003
4	EAT004	noah.jones@eat.app	Olivia Brown	Engineering	LAP1004
5	EAT005	ava.miller@eat.app	James Wilson	Marketing	LAP2001
6	EAT006	isabella.davis@eat.app	James Wilson	Marketing	LAP2002
7	EAT007	sophia.garcia@eat.app	James Wilson	Marketing	LAP2003
8	EAT008	mason.martinez@eat.app	James Wilson	Marketing	LAP2004
9	EAT009	ethan.lopez@eat.app	Michael Clark	Sales	LAP3001
10	EAT010	mia.gonzalez@eat.app	Michael Clark	Sales	LAP3002
11	EAT011	logan.wilson@eat.app	Michael Clark	Sales	LAP3003
12	EAT012	charlotte.anderson@eat.app	Michael Clark	Sales	LAP3004
13	EAT013	james.thomas@eat.app	Sarah Lee	HR	LAP4001
14	EAT014	amelia.moore@eat.app	Sarah Lee	HR	LAP4002
15	EAT015	benjamin.taylor@eat.app	Sarah Lee	HR	LAP4003
16	EAT016	evelyn.white@eat.app	Sarah Lee	HR	LAP4004
17	EAT017	lucas.harris@eat.app	David King	Finance	LAP5001
18	EAT018	harper.martin@eat.app	David King	Finance	LAP5002
19	EAT019	alexander.thompson@eat.app	David King	Finance	LAP5003
20	EAT020	ella.garcia@eat.app	David King	Finance	LAP5004
21	EAT021	henry.martinez@eat.app	Olivia Brown	Engineering	LAP1005
22	EAT022	grace.rodriguez@eat.app	Olivia Brown	Engineering	LAP1006
23	EAT023	jackson.lee@eat.app	Olivia Brown	Engineering	LAP1007
24	EAT024	victoria.walker@eat.app	Olivia Brown	Engineering	LAP1008
25	EAT025	sebastian.hall@eat.app	James Wilson	Marketing	LAP2005
26	EAT026	scarlett.allen@eat.app	James Wilson	Marketing	LAP2006
27	EAT027	aiden.young@eat.app	James Wilson	Marketing	LAP2007
28	EAT028	zoe.hernandez@eat.app	James Wilson	Marketing	LAP2008
29	EAT029	jack.king@eat.app	Michael Clark	Sales	LAP3005

	A	B	C	D	E	F
	Asset ID	OS	Model	Purchase Date	Deploy Date	Is Recycled
1	LAP1001	Windows 11	Dell XPS 13	2024-05-01	2024-05-15	No
2	LAP1002	Windows 11	MacBook Pro 14	2024-06-10	2024-06-25	No
3	LAP1003	Ubuntu 22.04	Dell Precision 5550	2023-12-20	2024-01-05	No
4	LAP1004	Windows 11	Lenovo ThinkPad X1	2024-03-15	2024-04-01	No
5	LAP1005	Windows 10	HP EliteBook 840	2023-09-05	2023-09-20	Yes
6	LAP1006	Ubuntu 20.04	Lenovo ThinkPad T14	2023-08-12	2023-08-28	Yes
7	LAP1007	Windows 11	Dell Latitude 7420	2024-01-22	2024-02-10	No
8	LAP1008	Windows 11	MacBook Air M2	2024-04-18	2024-05-02	No
9	LAP1009	Windows 11	Dell XPS 15	2024-07-07	2024-07-22	No
10	LAP1010	Windows 11	HP Spectre x360	2024-06-03	2024-06-17	No
11	LAP1011	Ubuntu 22.04	Dell Precision 3560	2024-02-14	2024-02-28	No
12	LAP1012	Windows 11	Lenovo Yoga 9i	2024-05-27	2024-06-10	No
13	LAP1013	Windows 10	HP EliteBook 850	2023-11-30	2023-12-15	Yes
14	LAP2001	Windows 11	MacBook Pro 16	2024-03-05	2024-03-18	No
15	LAP2002	Windows 11	Dell XPS 13	2024-04-10	2024-04-25	No
16	LAP2003	Ubuntu 22.04	Lenovo ThinkPad X1 Carbon	2024-01-05	2024-01-20	No
17	LAP2004	Windows 11	HP Elite Dragonfly	2024-02-23	2024-03-10	No
18	LAP2005	Windows 10	Dell Latitude 5420	2023-10-15	2023-10-30	Yes
19	LAP2006	Ubuntu 20.04	MacBook Air M1	2023-09-21	2023-10-05	Yes
20	LAP2007	Windows 11	Lenovo ThinkPad T15	2024-02-01	2024-02-14	No
21	LAP2008	Windows 11	HP Spectre x360	2024-05-12	2024-05-26	No
22	LAP3001	Windows 11	Dell XPS 13	2024-06-08	2024-06-23	No
23	LAP3002	Windows 11	MacBook Pro 14	2024-07-01	2024-07-15	No
24	LAP3003	Ubuntu 22.04	Dell Precision 5750	2024-03-28	2024-04-12	No
25	LAP3004	Windows 11	Lenovo ThinkPad X1 Yoga	2024-05-16	2024-05-30	No
26	LAP3005	Windows 10	HP EliteBook 840	2023-11-11	2023-11-26	Yes
27	LAP3006	Ubuntu 20.04	Lenovo ThinkPad T14	2023-10-09	2023-10-22	Yes
28	LAP3007	Windows 11	Dell Latitude 7420	2024-01-30	2024-02-14	No
29	LAP3008	Windows 11	MacBook Air M2	2024-04-21	2024-05-05	No

Motivation & Objectives



Reduce toil in IT operations and security



Improve employee experience and productivity



Provide logistical and financial insights for stakeholders

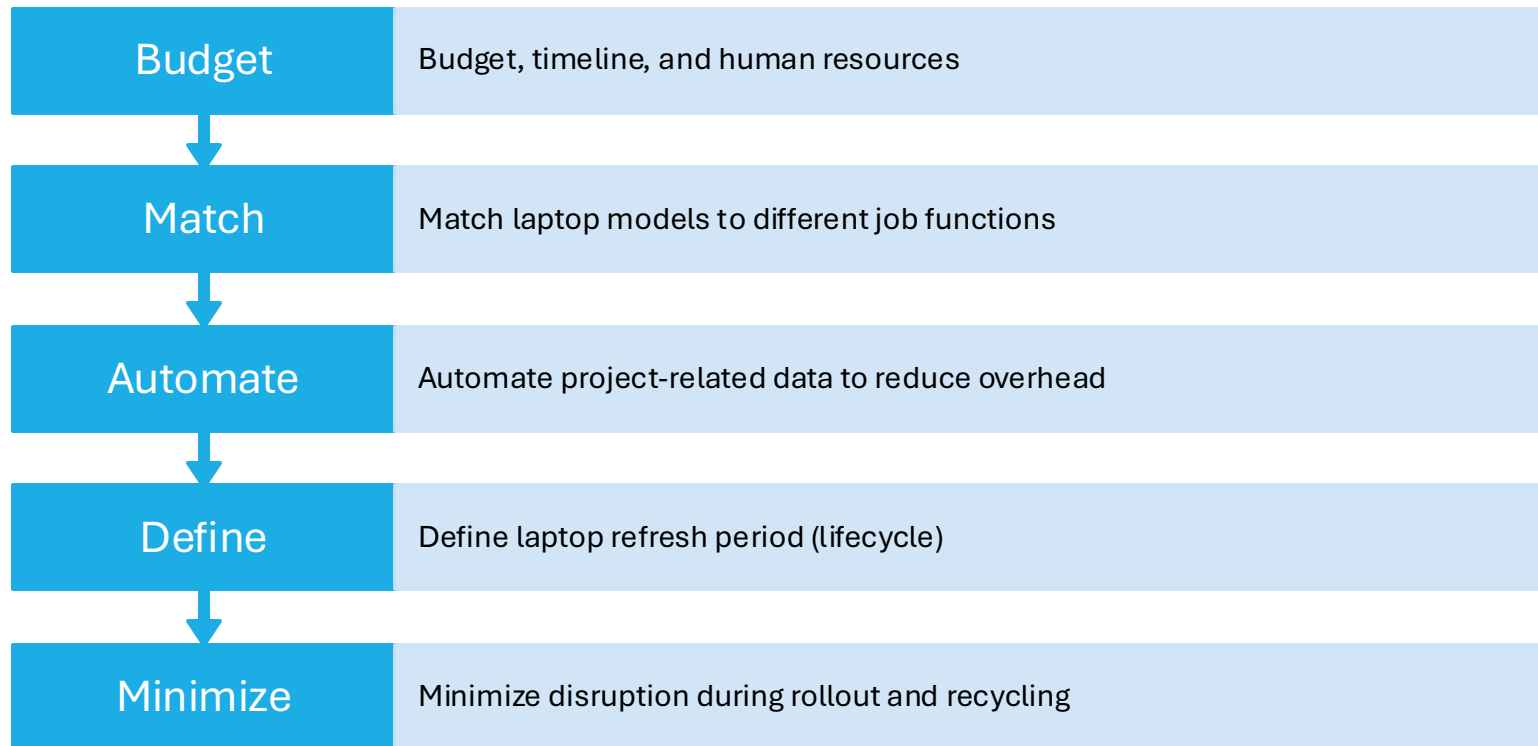


Integrate with existing tools and solutions in the business



Build a solution that can also be marketed externally

Scope & Key Considerations



Know thy enemies

Market Opportunity

- ▶ ITAM market $\approx$ US\$2.85B
- ▶ Internal benefit: reduced operational costs
- ▶ External benefit: new SaaS revenue stream
- ▶ Leverages existing asset and HR management expertise

Market & Positioning



Tailored internal ITAM + laptop refresh orchestration



Extensible as a SaaS platform



Competitors: IBM Maximo, ServiceNow, SolarWinds, Azure MDM



Gap: few products focus on periodic laptop refresh workflows

Money Talks

Budgets

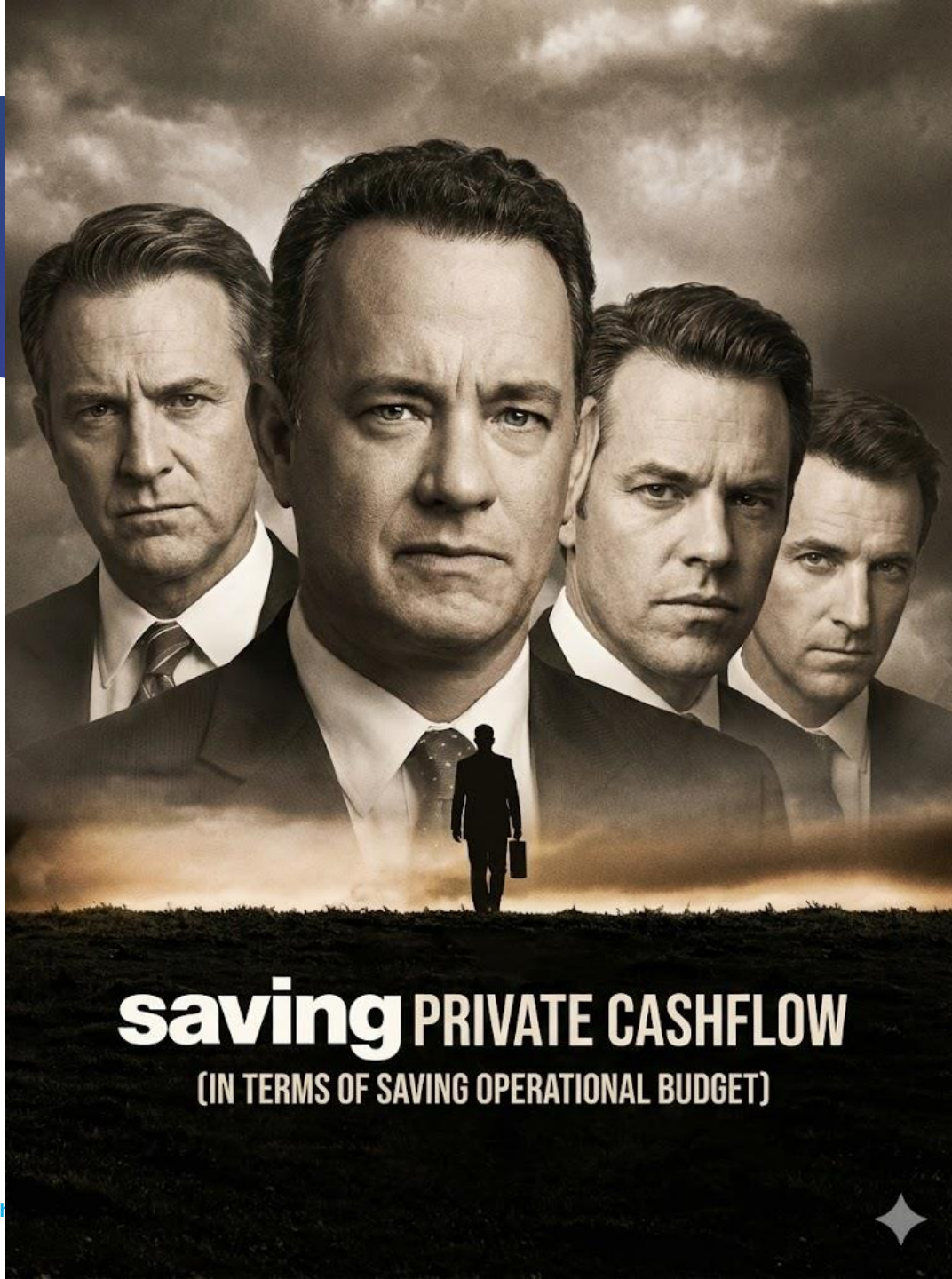
- ▶ Immediately earmarked **\$1M** (1% of annual net revenue) for the MVP stage of the product lifecycle
 - ▶ Research & Development
 - ▶ Infrastructure
 - ▶ Toolings + Licensing (if necessary)
 - ▶ Operational & Management
- ▶ Human Resources
 - ▶ 2 senior developers (\$10,000 monthly cost per person)
 - ▶ 2 mid-level (\$8,000 monthly cost per person)
 - ▶ Assign Product Owner (\$4,000 monthly cost, shared between products)
 - ▶ Assign Scrum Master (\$4,000 monthly cost, shared between teams)

Budgets (cont.)

- ▶ Reaching the MVP state in 5 months
 - ▶ Personnel cost: \$240,000 (base) + \$160,000 (OT, bonus) (24% + 16%)
 - ▶ Infrastructure cost: \$100,000 (10%)
 - ▶ Toolings & Licensing cost: \$100,000 (10%)
 - ▶ Operational & Management cost: \$300,000 (30%)
 - ▶ Budget Overhead: \$100,000 (10%)
- ▶ Additionally commit \$1M more
 - ▶ Needs evaluation, retrospection
 - ▶ Needs C-Suite approval
- ▶ Hard-capped at \$2.5M in total
 - ▶ Considered failed and deprecated when MVP is not satisfied at this point

Cashflow Estimations

- ▶ 5 months of development
 - ▶ \$1M - \$3M
- ▶ Operating at a loss (onboarding)
 - ▶ \$3M (capped)
- ▶ Return on Investment (ROI, first positive cashflow)
 - ▶ First 3 months after onboarding: \$200,000 Monthly Recurring Net Revenue (MRNR)
 - ▶ Next 3 months: \$400,000 (200% increase) MRNR, use this as stable assumption
 - ▶ Break-even point:
 - ▶ Best case: 8.5 months
 - ▶ Worst case: 16.5 months (1.25 years)
 - ▶ Note: Rough estimation, does not account for improved MRNR after first 6 months



saving PRIVATE CASHFLOW

(IN TERMS OF SAVING OPERATIONAL BUDGET)



Saving Private Cashflow

- ▶ Reduce IT personnel time and effort
 - ▶ ITOps dept has 20 headcount (as of now)
 - ▶ \$5,000 average monthly cost per headcount
 - ▶ \$1M monthly cost in total
 - ▶ 50% of that must be allocated when refreshing laptops
 - ▶ Saving \$500,000
- ▶ Reduce transitioning time and effort for end-users
 - ▶ Estimate to reduce from 1.5 days to 1 day (33.33% decrease) of work
- ▶ Less effort for the Business Intelligence and Financial Operations roles
 - ▶ Improved data retrieval, transformations and analysis
 - ▶ Richer and more accurate insights

“

The customers of my enemy is
my friend

”

SOME VERY OLD DUDES A LONG TIME AGO

Who are we selling to?

NOT ONLY DO WE SAVE MONEY OURSELVES

WE MAKE MORE MONEY IN THE PROCESS

Target Customers

- ▶ SMBs and large enterprises issuing laptops to employees
- ▶ Banking & Finance (e.g., Desjardins, BMO, RBC, CIBC, Scotia, AMEX)
- ▶ IT & Software companies
- ▶ Heavy industries, defense, government, contractors

Target Users / Roles



IT operations and support teams



Security teams (SOC analysts, InfoSec)



Auditors and GRC teams



Business, financial analysts and project managers

Solution Overview

- ▶ Web-based SaaS full-stack application
 - ▶ Native desktop application down the road
- ▶ Project-based laptop refresh and IT asset orchestrations
- ▶ Designed for internal use and external customers
- ▶ Multi-tenant, scalable and secure platform

High-Level Architecture



Service-oriented Architecture (SoA)



Monolithic frontend (initially) with scalable structure



Backend microservices (Asset, User, Auth, Admin)

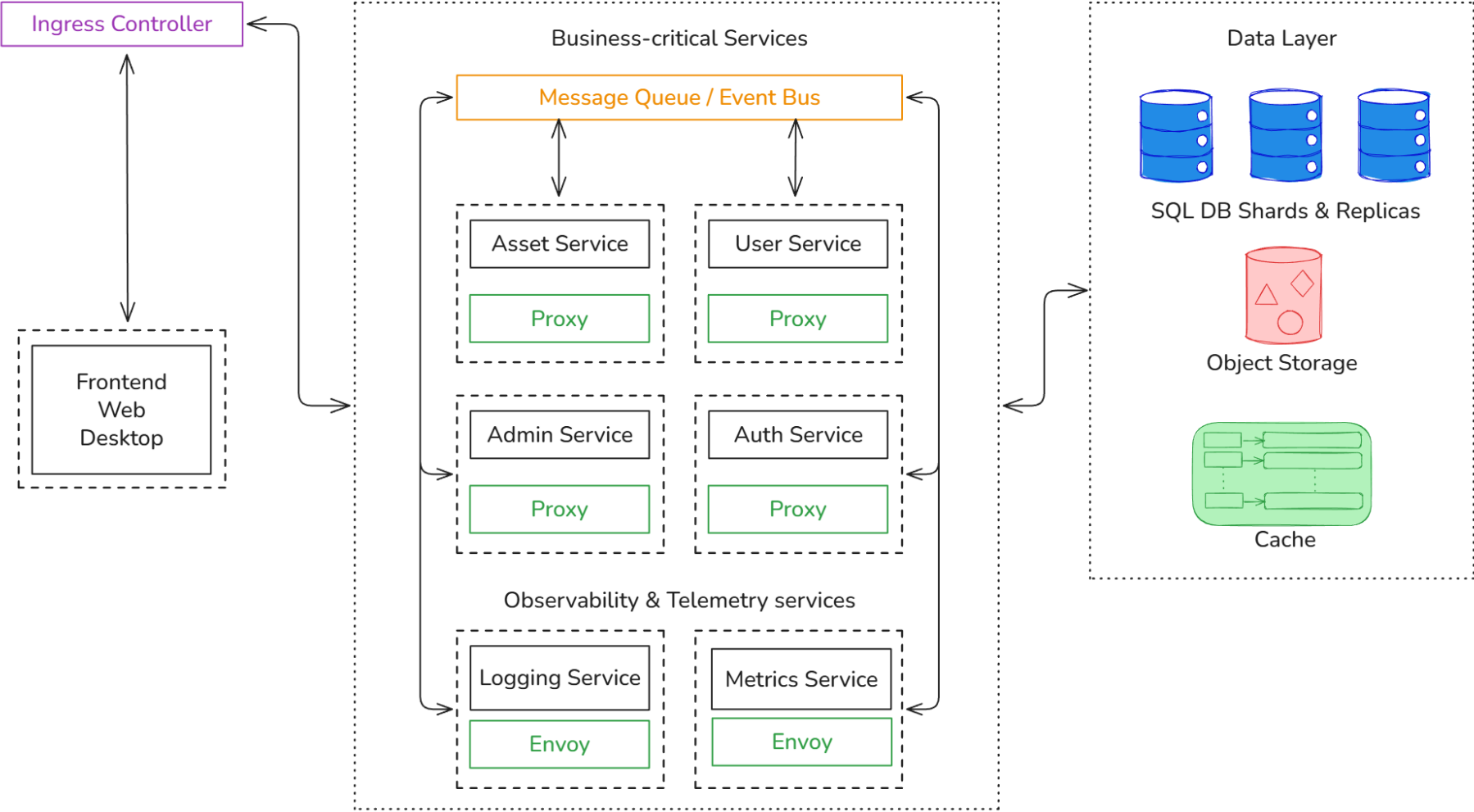


Service mesh + message broker for resilience

Cluster Architecture

Application Firewall

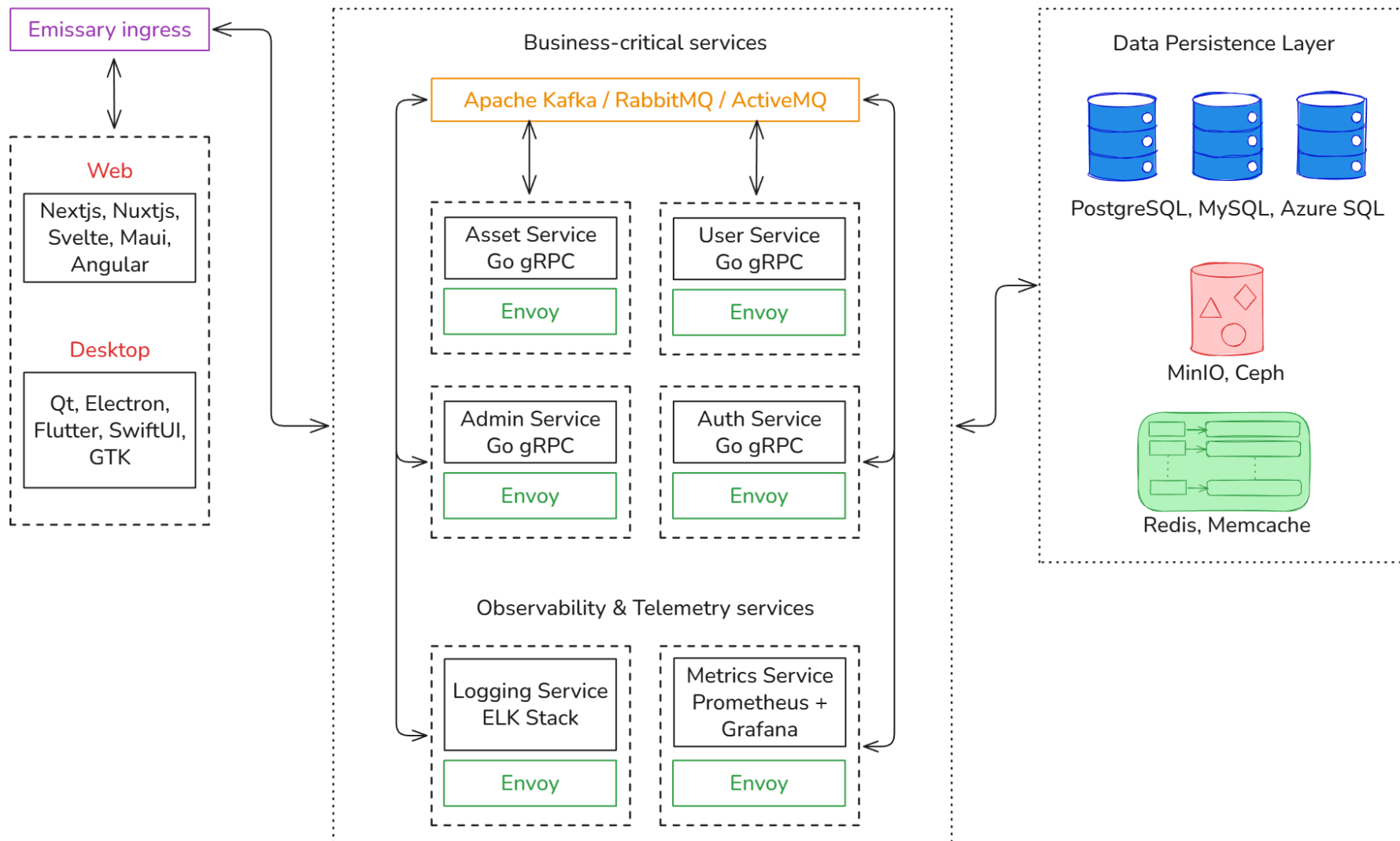
Service Mesh



Technology Stack (options)

Cloudflare (Zero Trust + WAF)

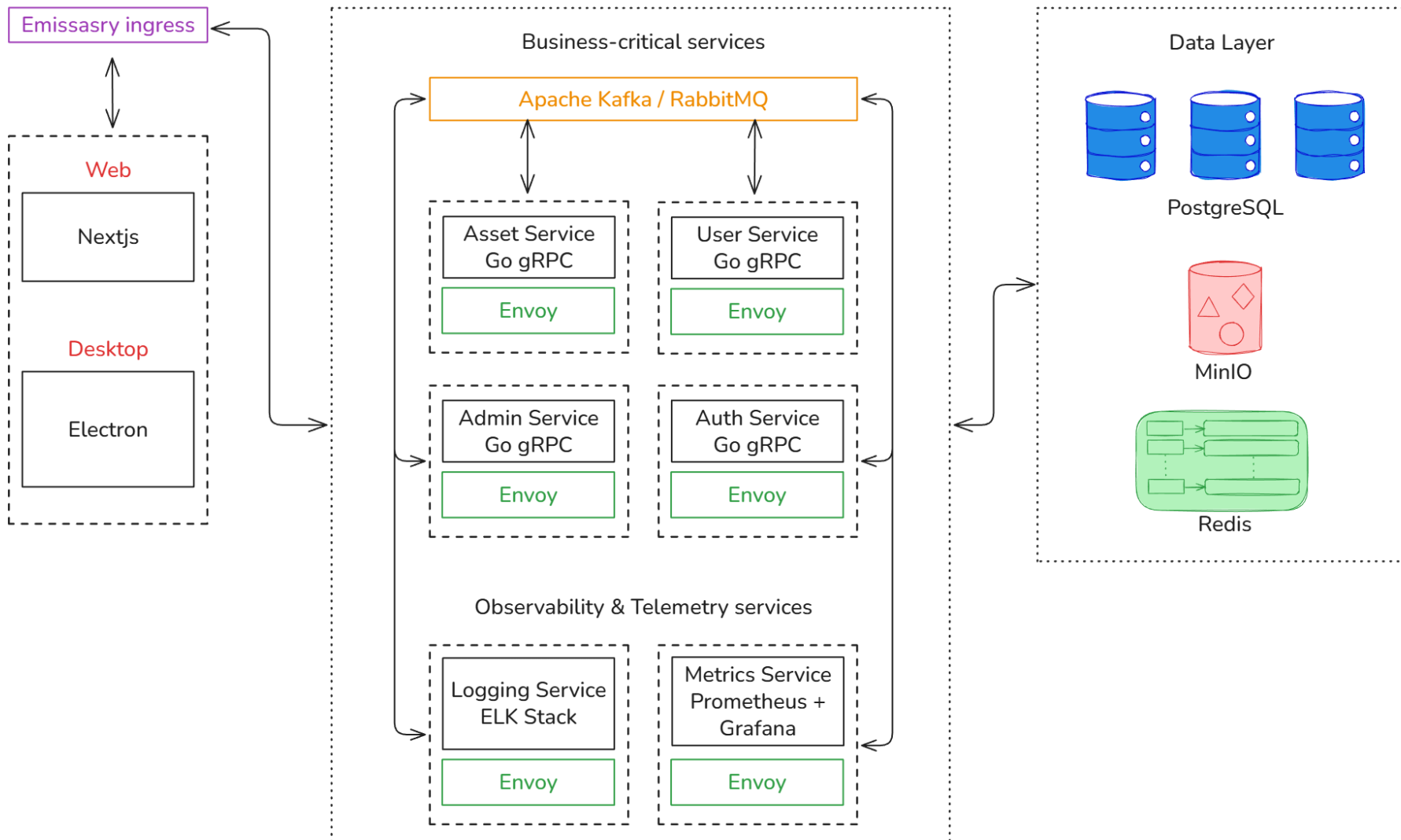
Consul, Istio Service Mesh



Technology Stack (decisions)

Cloudflare (Zero Trust + WAF)

Istio Service Mesh



Cloudflare (Zero Trust + WAF)

FREE

&

OPEN SOURCE

SOFTWARE

Emissary ingress

Web

Nextjs

Desktop

Asset Service
Go gRPC

Envoy

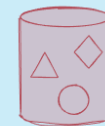
Service
gRPC

by

Data Layer



PostgreSQL



MinIO

Key Usage Workflows

- ▶ User accesses web or desktop client
- ▶ Organization-defined sign-in and AuthN/Z
- ▶ Redirected to Overview dashboard after login
- ▶ Navigate to asset, user, financial or upgrade/refresh dashboards
- ▶ User logs out to end session

Core Business Processes

- ▶ Asset management: view, add, remove, assign assets
- ▶ Disable, reset or quarantine assets as needed
- ▶ User management: add/remove users, sync with directories
- ▶ Financial and operational insights across assets

Upgrade / Refresh Process



Plan single or batch laptop upgrades



Onboard new assets and recycle old ones



Track statuses: Started → Preparing → Shipped → Delivered



Then: Received old asset → Recycled → Finished



Generate shipping labels for users

Assumptions, Risks & Constraints

- ▶ Assumptions: stable cloud infrastructure, teams adopt new processes
- ▶ Risks: rollout disruption (e.g. theft, dead on arrival), integration challenges, security concerns
- ▶ Constraints: budget, resources, compliance and regulations
- ▶ Contingency: phased rollout, pilots, rollback strategies

Development Methodology Options



Agile



Waterfall



Incremental



Each evaluated for fit with SaaS delivery

Methodologies – Pros & Cons



Agile: frequent feedback, early increments, continuous testing



Waterfall: clear requirements, predictable timeline, strong documentation



Incremental: functional slices, faster initial value, easier debugging

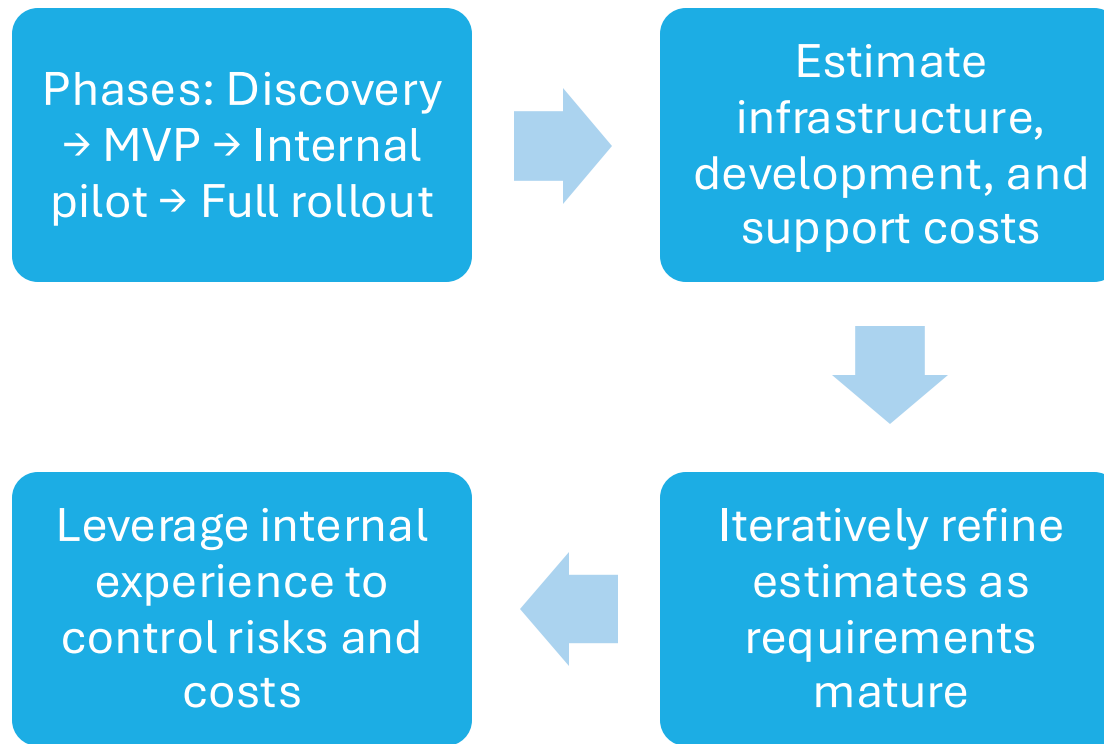


Each has trade-offs in flexibility, risk and integration complexity

Final Choice: Agile

- ▶ Short iterations and continuous delivery for SaaS
- ▶ Feature flags and gradual rollouts
- ▶ Telemetry-driven prioritization and validation
- ▶ Reduces release risk and keeps product responsive to users

Preliminary Timeframe & Cost



System Analysis & Design Overview

- ▶ Use case diagrams and activity diagrams
- ▶ Textual use case descriptions
- ▶ Class and sequence diagrams
- ▶ Preliminary product backlog with priorities and complexities

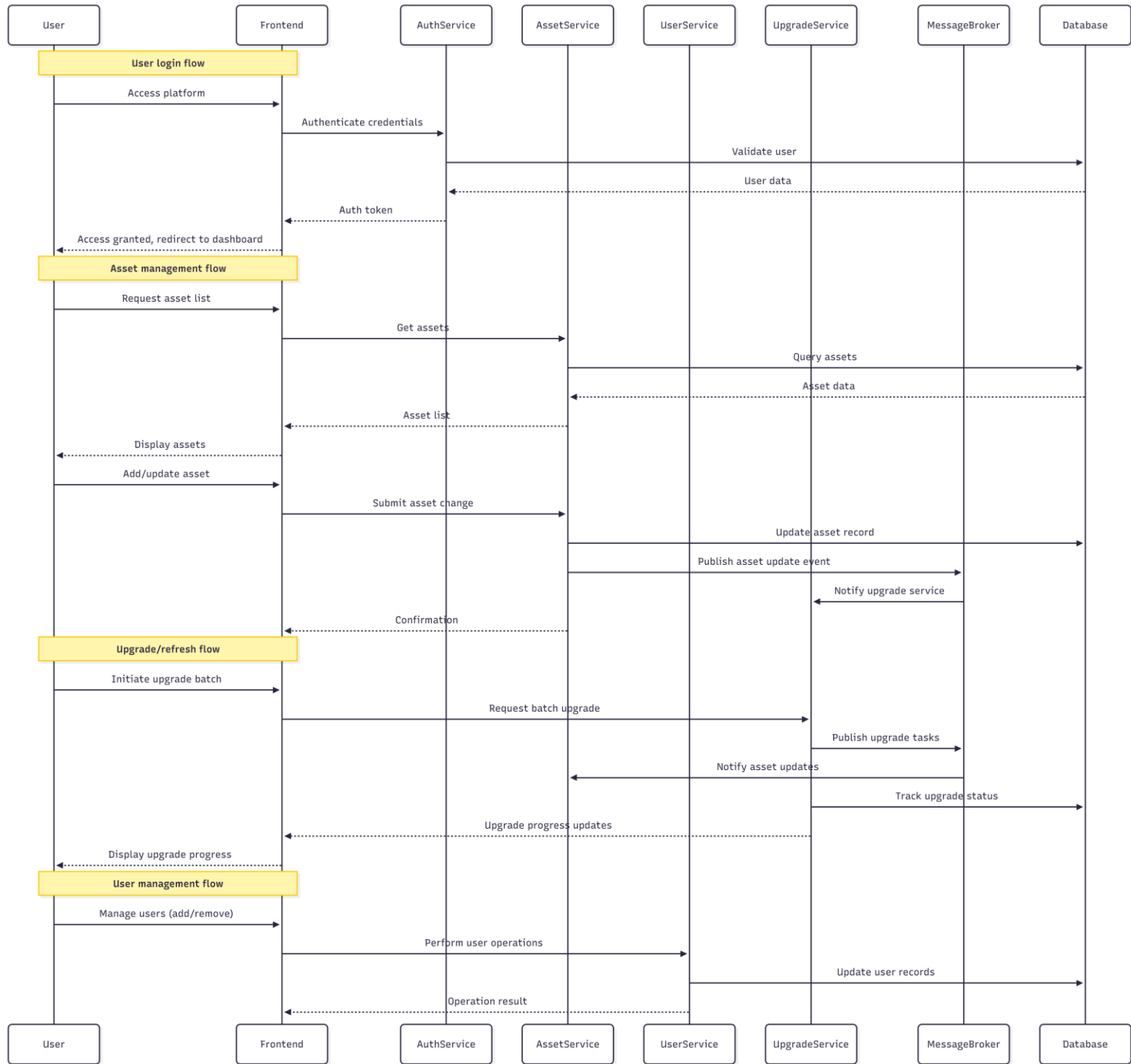
Textual Use Cases & Class Diagrams

- ▶ Example use case: Batch Laptop Refresh
- ▶ Actors: IT operator and system
- ▶ Main flow: select users → plan shipment → track progress → complete
- ▶ Class diagram entities: User, Asset, RefreshPlan, Shipment, Organization, Invoice

Sequence Diagram

Sequence Diagrams & Product Backlog

- ▶ Sequence example: Start Batch Upgrade (UI → API → Broker → Services → DB)
- ▶ Backlog items aligned with roles and workflows
- ▶ Each item has priority and complexity estimates
- ▶ Supports Agile planning and incremental delivery



Preliminary Product Backlog

[HTTPS://AAANH.ATLASSIAN.NET/JIRA/SOFTWARE/PROJECTS/LRI/BOARDS/2/TIMELINE?TIMELINE=MONTHS](https://aaanh.atlassian.net/jira/software/projects/LRI/boards/2/timeline?timeline=months)

Work	January '26	February '26	March '26	April '26	May '26
Sprints					
[LR]-47 Components Scaffold... [LR]-55 Initialize web frontend stack [LR]-56 Initialize auth service [LR]-57 Initialize asset servi... [LR]-58 Initialize user service [LR]-59 Initialize admin service [LR]-60 Create initial database configuratio... [LR]-61 Create initial message queue configurati... [LR]-70 Create proxy configuratio...					
[LR]-48 Frontend MVP [LR]-8 Display asset compliance chart [LR]-12 Add asset UI for... [LR]-19 User list UI [LR]-40 Session management caching [LR]-1 User login UI [LR]-5 Logout functionality [LR]-21 Add user UI form [LR]-6 Dashboard overview UI [LR]-9 Mobile responsive dashboard [LR]-10 Asset list UI [LR]-14 Update asset UI [LR]-17 Assign asset to user UI [LR]-25 Batch upgrade UI [LR]-26 Batch upgrade A... [LR]-28 Single asset upgrade... [LR]-30 Print shipping label...					
[LR]-49 Backend MVP [LR]-20 Get users API [LR]-23 Remove user API [LR]-29 Single asset upgrade A... [LR]-32 Recycle asset workfl... [LR]-7 Fetch dashboard metrics A... [LR]-31 Generate shipping label A... [LR]-34 Message publish from servic... [LR]-1 User login backend [LR]-3 JWT token generation [LR]-4 Role-based access control [LR]-11 Get assets API [LR]-13 Create asset API [LR]-15 Update asset API [LR]-16 Delete asset API [LR]-18 Assign asset API [LR]-27 Track upgrade status A... [LR]-35 Message consume in services [LR]-22 Create user API					
[LR]-52 Database MVP [LR]-37 Database schema users [LR]-39 Cache setup [LR]-36 Database schema asse... [LR]-38 Database schema upgrad... [LR]-24 Rsync directory users					
[LR]-50 Observability Developme... [LR]-51 Infrastructure MVP [LR]-41 Service mesh installation [LR]-33 Message broker setup [LR]-42 Configure service mesh routing [LR]-72 Ingress Controller setup [LR]-73 Cloudflare Zero Trust + WAF set...					
[LR]-53 DevSecOps & CI/CD Set... [LR]-45 CI/CD pipeline set... [LR]-46 Rolling upgrades and rollba...					
[LR]-54 Onboarding [LR]-71 Security [LR]-44 Encrypt data at re... [LR]-43 Encrypt data in tran...					